

► BACKGROUND

Dementia causes progressive cognitive decline and early dependence on caregivers, who often face significant physical and emotional challenges [1]. With an aging global population, the prevalence of dementia continues to rise. Therefore, monitoring systems that provide real-time data can help caregivers deliver more responsive and personalized care [2].

► METHODOLOGY

As part of a wider system represented in Figure 1 under development, two wearable devices are being developed: and In-ear Wearable Device (HowMI) [3], and a Wrist Wearable Device [4]. This paper focuses on the Wrist Device, which has functional requirements aimed at continuously monitoring heart rate, blood oxygen saturation, step count, sleep patterns, real-time location, falls, and emergency situations. Its development is also guided by four domains of non-functional requirements: physical, emotional, cognitive, and implementation success.

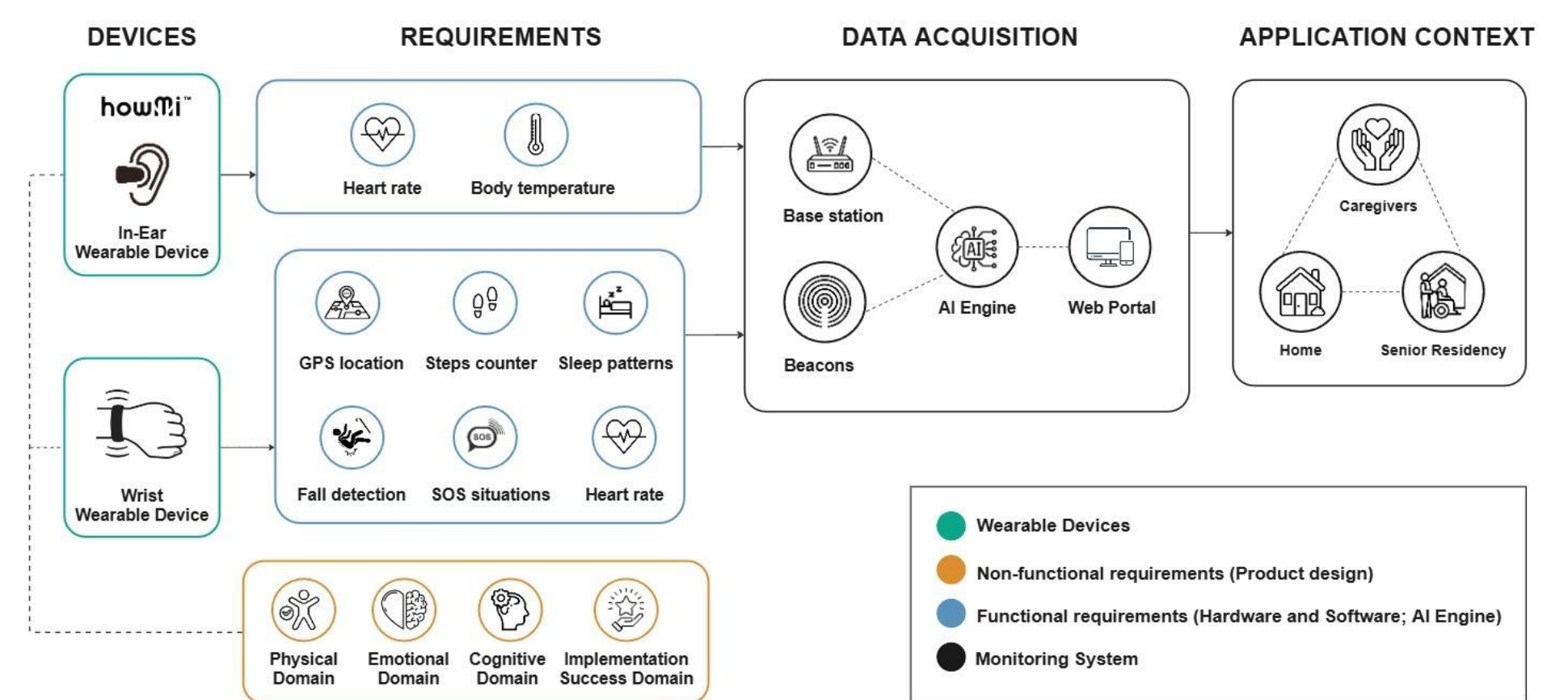


Figure 1. Overview of the Project System

► HARDWARE AND SOFTWARE

A functional prototype of the wearable device was developed using development boards for the selected sensors: BMI270 for motion tracking, SAM-M10Q for outdoor geolocation and time synchronization, and MAX30101 for heart rate and blood oxygen saturation measurement. Data acquisition is managed by a real-time operating system (RTOS), with secure data transmission via encrypted BLE.

► FUTURE WORK

- Analyzing the device's current model to align its technology and design, considering wrist anthropometry to create a final model suitable for the elderly.
- Key elements like buttons and straps will be studied for usability, with textures and colors differentiating button functions, and a safety system added to the strap to prevent accidental removal.
- An extensive dataset for monitoring dementia, will be created, integrating advanced AI techniques to improve real-time monitoring.
- Efforts will prioritize optimizing data acquisition algorithms, reducing power consumption, and enhancing communication protocols, including testing LoRa (Long Range) with AES-CTR (Advanced Encryption Standard - Counter) encryption for security.
- A hardware redesign will integrate sensors on a compact PCB (Printed Circuit Board).
- Software improvements will add features, optimize data processing, and enhance energy efficiency, aiming to deliver a cost-effective, user-beneficial product

► USABILITY AND ERGONOMIC DESIGN

Several 3D models and test prototypes were developed to optimize the sensor layout and overall device dimensions, addressing constraints such as mechanical tolerances and space allocation for connectors. A functional model was achieved, Figure 2. The final rectangular form factor balances comfort and component arrangement. A smaller screen size was also chosen to match the anatomy of the adult wrist, contributing to a more compact and comfortable device.

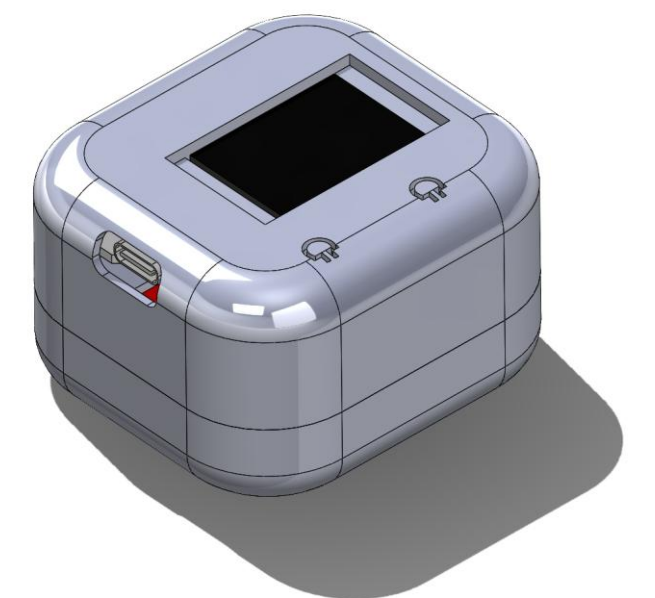


Figure 2. 3D representation of the final functional model

► ACTIVITIES RECOGNITION

AI models, such as, SVM, RF, KNN, XGBoost, LightGBM, CNN, LSTM, and GRU were trained on public datasets for activity classification including walking, sleeping, eating, and hygiene-related tasks.

This study also explored the application of unsupervised learning approaches to detect deviations from users' typical daily routines using GRU models trained on synthetic normal data. Leveraging autoencoders and sequence-to-sequence models with attention mechanisms. These models detect deviations in daily routines through reconstruction errors and prediction discrepancies, achieving 99% anomaly recall.

► BIBLIOGRAPHY

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